

Customer-Constrained Optimization for Personalized Mortgage Rate Pricing

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Abstract—This paper presents a customer-constrained optimization framework for personalized mortgage rate pricing in retail banking. The objective is to recommend the lowest feasible mortgage rate for each application while preserving near-optimal expected profit. The framework integrates borrower-level features, acceptance modeling, and default risk estimation within a unified expected profit formulation.

Using a Freddie Mac-derived loan-level dataset, we demonstrate that the proposed approach achieves strong predictive performance ($AUC \approx 0.79$) and consistent optimization behavior across a heterogeneous borrower population. Empirical results show that the customer-constrained strategy reduces average mortgage rates by approximately 12.5 basis points, while incurring only a 3.97% reduction in expected profit.

These findings suggest that modest, data-driven adjustments to pricing can significantly improve customer outcomes while maintaining profitability. The framework provides a practical and interpretable approach for implementing customer-centric pricing strategies in competitive lending environments.

Index Terms—mortgage optimization, personalized pricing, credit risk, profit maximization, constrained optimization

I. INTRODUCTION

Mortgage pricing has traditionally relied on segmented approaches based on borrower characteristics. However, advances in machine learning have enabled more granular credit risk modeling and pricing optimization [2], [5].

This paper proposes a framework that determines the lowest feasible mortgage rate for each application while maintaining near-optimal expected profit.

II. MORTGAGE APPLICATION INPUTS AND ASSUMPTIONS

Each mortgage application is represented by a feature vector:

$$x = (\text{credit score, income, DTI, LTV, loan amount, } \dots) \quad (1)$$

The model assumes:

- Credit score reflects borrower creditworthiness
- Income and debt-to-income (DTI) capture repayment capacity
- Loan-to-value (LTV) captures collateral risk
- Loan amount L directly impacts exposure

The cost of funds $c(x)$ may depend on borrower risk. Loss given default $LGD(x)$ is assumed to increase with LTV.

All optimization is performed at the individual application level.

III. PROBLEM FORMULATION

We formulate the mortgage pricing problem at the individual application level, where all quantities are defined with respect to a given borrower feature vector x and offered rate r . The feasible mortgage rate set is:

$$\mathcal{R} = [0.03, 0.065] \quad (2)$$

We define expected profit as the product of acceptance probability and conditional profit:

$$\Pi_{\text{exp}}(x, r) = P(A = 1 \mid x, r) \cdot \Pi(x, r) \quad (3)$$

where $P(A = 1 \mid x, r)$ denotes acceptance probability and $\Pi(x, r)$ denotes conditional profit.

A. Notation and Definitions

We summarize key variables used in the formulation:

- x : Feature vector representing a mortgage application
- r : Offered mortgage interest rate
- $\mathcal{R}$: Feasible set of mortgage rates
- $P(A = 1 \mid x, r)$: Acceptance probability
- $P_d(x, r)$: Default probability
- $\Pi_{\text{exp}}(x, r)$: Expected profit
- $\Pi^*(x)$: Maximum achievable expected profit
- $r^*(x)$: Customer-constrained optimal rate
- ε : Profit tolerance parameter

IV. PROFIT MODEL

We define the profit from a mortgage loan as a function of the offered rate and borrower characteristics. The formulation captures interest income, expected credit losses, and acquisition costs.

$$\Pi(x, r) = L(r - c(x))T - P_d(x, r) LGD(x) L - C_{\text{acq}}(x) \quad (4)$$

where L denotes loan amount and T denotes loan term.

A. Economic Interpretation of the Profit Function

The expected profit formulation captures three competing forces in mortgage pricing:

- **Interest Margin:** Higher mortgage rates increase per-loan revenue through the spread $(r - c(x))$.
- **Acceptance Behavior:** As rates increase, borrower acceptance probability $P(A = 1 \mid x, r)$ declines, reducing loan origination volume.

- **Default Risk:** Higher rates may also increase borrower stress, leading to higher default probability $P_d(x, r)$.

These competing effects naturally induce a unimodal relationship between expected profit and mortgage rate, where profit increases initially with rate due to margin expansion, and then decreases due to declining acceptance and increasing default risk.

The trade-off between pricing and demand is well studied in revenue optimization literature [7].

V. BEHAVIORAL MODELS

We model borrower acceptance and default behavior as functions of both borrower characteristics and the offered mortgage rate.

The acceptance probability is defined as:

$$P(A = 1 \mid x, r) = \frac{1}{1 + \exp(-(\beta_0 + \beta^\top x + \gamma r))} \quad (5)$$

This functional form ensures that probabilities remain bounded between 0 and 1 while capturing nonlinear sensitivity to the offered rate.

The default probability is modeled as:

$$P_d(x, r) = \frac{1}{1 + \exp(-(\alpha_0 + \alpha^\top x + \eta r))} \quad (6)$$

A. Modeling Assumptions

The acceptance and default models are specified using logistic regression forms to capture nonlinear dependencies on borrower characteristics and mortgage rate.

- The acceptance model $P(A = 1 \mid x, r)$ captures borrower sensitivity to pricing, where higher rates reduce the likelihood of loan acceptance.
- The default model $P_d(x, r)$ captures credit risk, where borrower characteristics and pricing jointly influence repayment behavior.

While these models can be replaced by more sophisticated machine learning models (e.g., gradient boosting), the logistic specification provides interpretability and analytical tractability.

VI. CUSTOMER-CONSTRAINED OPTIMIZATION

We define the profit-maximizing rate as the rate that maximizes expected profit over the feasible set:

$$\Pi^*(x) = \max_{r \in \mathcal{R}} \Pi_{\text{exp}}(x, r) \quad (7)$$

$$r^*(x) = \min \{r \in \mathcal{R} : \Pi_{\text{exp}}(x, r) \geq (1 - \varepsilon)\Pi^*(x)\} \quad (8)$$

A. Interpretation of Customer-Constrained Optimization

The optimal rate $r^*(x)$ maximizes expected profit for a given application. However, this purely profit-driven strategy may lead to higher pricing that is not competitive or customer-friendly.

To address this, we introduce a constraint that allows for a small tolerance ε in expected profit. The constrained rate is defined as the lowest rate that achieves at least $(1 - \varepsilon)$ of the optimal profit.

This formulation reflects a practical business objective: reduce pricing where possible, while ensuring that the loss in profitability remains bounded.

From a decision-making perspective, this can be viewed as a form of *controlled revenue sacrifice* to improve customer outcomes, competitiveness, and potential long-term retention.

VII. SIMULATION STUDY

We evaluate the proposed framework using a Freddie Mac-derived mortgage dataset with heterogeneous borrower and loan characteristics.

A. Dataset and Empirical Setup

Experiments are conducted using a publicly available dataset derived from Freddie Mac loan-level data, containing borrower attributes, loan characteristics, and delinquency outcomes.

Default is defined as loans reaching 90 or more days past due ($\text{MonthsDelinquent} \geq 3$), consistent with industry standards.

Borrower features include credit score, debt-to-income ratio, loan-to-value ratio, loan amount, servicing characteristics, and property attributes. Additional feature engineering includes nonlinear transformations and interaction terms.

Since real-world datasets do not provide counterfactual pricing outcomes, borrower acceptance behavior is modeled as a function of mortgage rate.

B. Evaluation Metrics

We evaluate the framework using the following metrics:

- **Average Expected Profit:** Mean of $\Pi_{\text{exp}}(x, r)$ across all applications
- **Average Selected Rate:** Mean of recommended rates
- **Profit Retention:**

$$\frac{\Pi_{\text{exp}}(x, r^*(x))}{\Pi^*(x)}$$

- **Acceptance Rate:** Average $P(A = 1 \mid x, r)$

C. Optimization Procedure

For each application x :

- 1) Generate candidate rates $r \in \mathcal{R}$
- 2) Compute:
 - $P(A = 1 \mid x, r)$
 - $P_d(x, r)$
 - $\Pi(x, r)$
 - $\Pi_{\text{exp}}(x, r)$

- 3) Compute optimal profit $\Pi^*(x)$
- 4) Select lowest rate $r^*(x)$ satisfying:

$$\Pi_{\text{exp}}(x, r) \geq (1 - \varepsilon)\Pi^*(x)$$

D. Evaluation

For each application:

- Evaluate a discretized grid of candidate rates
- Compute expected profit
- Compare pure-profit vs constrained rate

VIII. RESULTS

We present empirical results demonstrating both predictive performance and optimization behavior of the proposed framework.

A. Predictive Model Performance

Gradient boosting methods such as XGBoost have demonstrated strong performance in credit risk modeling tasks [5]. A gradient boosting model (XGBoost) is used to estimate default probability based on borrower, loan, and servicing features. The model achieves an AUC of approximately 0.79 on held-out data, indicating strong predictive performance for loan-level risk estimation.

B. Profit vs Rate Behavior

Expected profit exhibits a unimodal relationship with respect to the mortgage rate, reflecting the trade-off between margin expansion and declining acceptance probability.

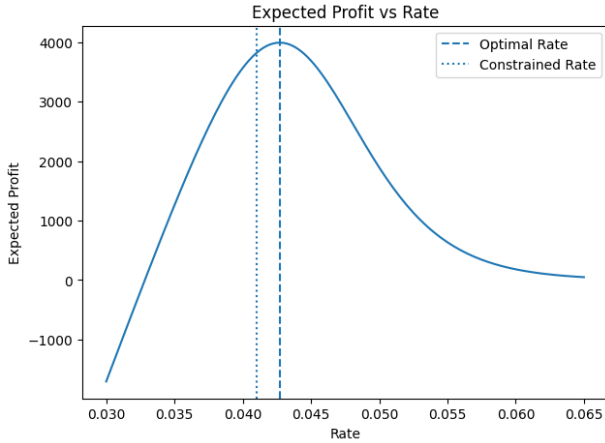


Fig. 1. Expected profit as a function of mortgage rate for a sample applicant

C. Aggregate Profit Behavior

To assess population-level behavior, we evaluate expected profit across 1,000 sampled applications.

The aggregate curve exhibits a clear unimodal structure, indicating a well-defined optimal pricing region. While individual applications display heterogeneity, the overall population behavior remains stable and predictable.

This suggests that pricing strategies can be designed using population-level insights while still allowing for individual-level personalization.

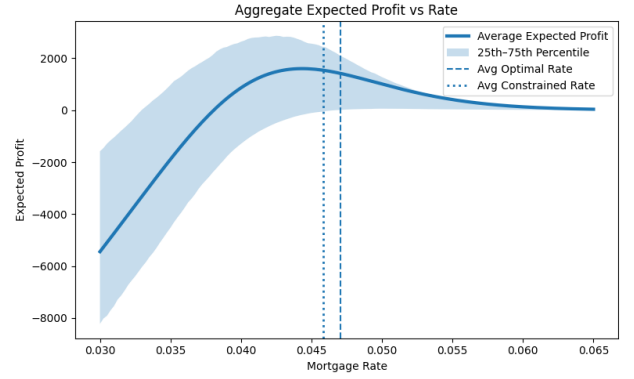


Fig. 2. Aggregate expected profit as a function of mortgage rate across 1,000 applications. The solid line represents mean expected profit, while the shaded region represents the interquartile range across borrowers.

D. Impact of Customer-Constrained Optimization

Across 1,000 sampled applications:

- Average optimal rate: 4.71%
- Average constrained rate: 4.58%
- Average reduction: 12.48 basis points
- Average profit reduction: 3.97%

These results demonstrate that customer-friendly pricing can be achieved with only a 3.97% reduction in expected profit, while reducing rates by approximately 12.5 basis points.

E. Rate Distribution Effects

The distribution of customer-constrained rates is consistently shifted toward lower values relative to profit-optimal rates, while maintaining substantial overlap.

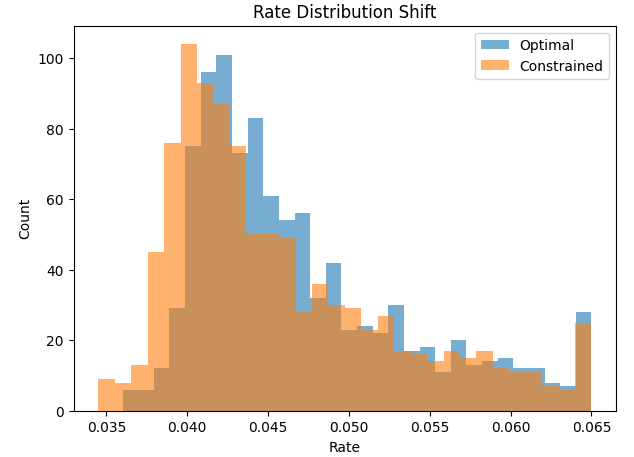


Fig. 3. Distribution of optimal vs customer-constrained rates

F. Robustness to Default Definitions

We evaluate multiple definitions of default based on delinquency duration thresholds (e.g., ≥ 3 , ≥ 5 , ≥ 10 months). As expected, predictive performance improves for stricter definitions due to clearer class separation, while the qualitative behavior of the optimization framework remains consistent.

G. Personalization Effects

Personalization primarily benefits lower-risk applicants, who receive reduced rates under constrained optimization, while higher-risk applicants exhibit minimal deviation from profit-optimal rates. This preserves risk-adjusted profitability while improving customer outcomes.

IX. MANAGERIAL IMPLICATIONS

The proposed framework provides a systematic approach to balancing profitability and customer competitiveness in mortgage pricing. By explicitly modeling the trade-off between rate, acceptance, and default risk, financial institutions can move beyond heuristic pricing strategies toward data-driven, application-level decision making.

The customer-constrained formulation offers a practical mechanism to improve customer outcomes while maintaining strong financial performance, making it particularly relevant in competitive lending environments.

X. CONCLUSION

This paper presents a customer-constrained mortgage rate optimization framework that balances profitability with customer competitiveness at the individual application level. By combining predictive modeling with constrained optimization, the framework identifies the lowest feasible rate that preserves near-optimal expected profit.

Using a Freddie Mac-derived loan-level dataset, the proposed approach demonstrates strong predictive performance ($AUC \approx 0.79$) and consistent optimization behavior across a heterogeneous population of borrowers. Empirical results show that the customer-constrained strategy reduces average mortgage rates by approximately 12.5 basis points while incurring only a 3.97% reduction in expected profit. These findings suggest that modest adjustments to pricing strategies can significantly improve customer outcomes with minimal impact on profitability.

The results also highlight the importance of incorporating borrower behavior into pricing decisions. In particular, the dependence of both acceptance probability and default risk on the offered rate introduces a natural trade-off that is not captured in traditional pricing frameworks.

This work has several practical implications for retail banking. The proposed framework can be integrated into existing pricing systems to enable personalized, data-driven rate recommendations that align with both business objectives and customer experience goals.

Several limitations remain. The acceptance model is specified rather than directly estimated from market data, and macroeconomic factors are not explicitly incorporated. Future work may extend the framework by incorporating competitive market dynamics, richer borrower behavioral signals, and dynamic pricing strategies over the loan lifecycle.

This suggests that traditional pricing approaches may leave measurable efficiency gains unrealized.

Overall, the proposed approach provides a simple yet effective framework for customer-centric mortgage pricing and

highlights the potential of combining machine learning with optimization in financial decision-making.

This suggests that customer-centric pricing strategies can be systematically implemented without sacrificing core financial performance.

XI. REPRODUCIBILITY AND CODE AVAILABILITY

To support reproducibility, the implementation of the proposed framework, including data preprocessing, model training, and optimization routines, is made publicly available.

The code repository includes:

- Data preprocessing and feature engineering pipelines
- Default prediction model (XGBoost)
- Mortgage rate optimization framework
- Scripts for generating all figures in this paper

The code can be accessed at:

<https://github.com/srikanth421/mortgage-rate-optimization>

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